



Towards a machine-interpretable calibration certificate for radioactivity

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ABSTRACT

Digitalisation of calibration laboratories, research institutes, and manufacturing facilities enables fast, automated flow of data and information, improving productivity while reducing turnaround times and error rates. Digital, machine interpretable calibration certificates are slated to replace or augment existing paper- or pdf-based certificates and will seamlessly integrate into digital workflows without the need for manual transcription of calibration information. National and international harmonisation of digital calibration information is crucial for their utilisation in the calibration community.

In this contribution, we will present digital, machine-interpretable calibration certificates for measurements of radioactivity and related quantities such as activity concentration and activity ratios. These certificates implement the XML-DCC (Digital Calibration Certificate in eXtensible Markup Language) schema developed at PTB. Descriptions of the calibration items, measurement methods, detected nuclides, and activity measurements are represented in a machine-interpretable way, employing a digital representation of the SI. We will discuss how such calibration certificates may be harmonised nationally and internationally, and show different use cases, from laboratory calibrations to key comparisons. Human-readable representations of these certificates, necessary for quality assurance and legacy processes, are discussed as well.

1. Introduction

1.1. Digital calibration certificates

In the age of digitisation and industry 4.0, it is clear that paper-based calibration certificates are no longer sustainable. With an increasing number of sensors and devices requiring calibration, the printing, mailing, storing, and processing of paper calibration certificates not only creates a bottleneck, but also increases the chance for errors and mistakes to happen at each step of the chain. Electronic (digital) calibration certificates mitigate these issues. We have to distinguish between different levels of “digital-ness”. In this contribution, we will focus on calibration certificates of level three or higher according to the extended utility model of Czarny et al. (2021) - that is, certificates with machine readable (level 3), machine interpretable (level 4) and/or machine executable (level 5) contents.

Multiple formats for digital calibration certificates and similar documents have been proposed, including some that attach machine interpretable data to human readable PDF documents (cf. e.g. Boschung et al. (2021), Coulon et al. (2023)). Other approaches focus on having a machine interpretable document such as XML (Extensible Markup Language) only (Hackel et al., 2023). Advantages of using pure XML

over PDF-with-XML (or PDF-with-JSON: Portable Document Format, JavaScript Object Notation) include no need for additional steps to extract the machine readable data from the PDF as well as no potential for a mismatch between human readable and machine readable document. On the other hand, XML certificates require additional information to allow customers to extract information in a meaningful way, for example in the form of XSLT (eXtensible Stylesheet Language Transformations) style files, which contain instructions for web browsers to display XML files. These would have to be distributed with the calibration certificate or hosted online.

We have chosen to implement calibration certificates in the XML format developed and hosted at PTB (Physikalisch-Technische Bundesanstalt, National Metrology Institute), which will be described in Section 1.2.

1.2. The PTB DCC

In this document, the abbreviation “DCC” will refer specifically to the XML-based format for digital calibration certificates developed at PTB (Hackel et al., 2021, 2023). This DCC is based on an XML schema in XSD format, which is maintained and hosted by PTB.² The

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² See www.ptb.de/dcc/ for more information and details.

current release is version 3.3.0.³ The schema is complemented by a vocabulary database,⁴ good practice examples,⁵ and usage guides (e.g., Foyer et al. (2024), Bartl et al. (2024)). The vocabulary database and usage guides are developed and maintained by technical committees (TCs) within DKD,⁶ the German Calibration Service at PTB.

XSLT style sheets may be used to render XML DCCs for human readability (Demir et al., 2022). External files, including human readable representation of the document, may also be included in the DCC (where allowed by the schema) as base 64 binary strings.

The DCC schema depends on the D-SI schema (Physikalisch-Technische Bundesanstalt, 2024; Hutzschenreuter et al., 2020) for the representation of physical quantities such as measurement results and on a modified version of the W3C's xmldsig schema⁷ for digital signatures, and otherwise stands alone. Further harmonisation and modularisation of the digital quality infrastructure have been proposed, which would see the introduction of a general schema “DX”, upon which the schemata for the DCC and other documents, such as Digital Reference Materials or Digital Test Certificates could build (Öktem et al., 2022).

Each DCC document contains four sections, plus optional signature(s):

- **Administrative data** (mandatory): Information about the calibrating laboratory, customer, and calibration item, such as a device or artefact.
- **Measurement results** (mandatory): Results from calibration measurements, with optional meta data. The results can be quantitative (physical quantities) and/or qualitative (conformity pass/-fail).
- **Comments** (optional): Unregulated section for further information desired by the customer or calibration laboratory.
- **Document** (optional): Human readable representation (e.g. PDF) of the DCC, or other attached documents.
- **Signature(s)** (optional): Digital signature verifying the integrity of the DCC.

Many elements of the DCC may carry the optional “refType” attribute. RefTypes allow information to be stored and extracted in a harmonised way, making DCCs truly machine interpretable. To that end, each calibration community maintains a list of community-specific refTypes with definitions and usage notes. Currently, PTB hosts a database of refTypes.⁸ RefTypes consist of a prefix and base, separated by an underscore. The prefix denotes the domain or metrological area, with the prefix “basic” reserved for refTypes used by multiple communities. Only the combination of prefix and base uniquely identifies a refType — the same base may be used in different communities with different meanings or in different contexts. Where refTypes are allowed, an element may carry zero, one, or multiple refTypes.

Future versions of the DCC schema⁹ will support definitions of measurands in external vocabulary data bases, reducing the need for measurand-specific refTypes.

There are no general rules governing the implementation of new refType prefixes. New refTypes and prefixes are implemented by consensus in the relevant TC. RefType prefixes are based on existing categories such as the DKD TCs or BIPM service categories¹⁰ and community needs.

Many elements of the DCC may carry optional attributes id and/or refId. The value of id must be unique within each DCC. The refId attribute is then used to refer back to the element with a given id elsewhere in the DCC. This formalism can be used to associate (for example) a measurement result to the calibration item it corresponds to.

Many elements of the DCC have the child elements name (mandatory) and description. These are used to provide a (short) name and (more detailed) description of the element for human-readable representations of the DCC. Both name and description elements have at least one content child element, containing said name and/or description. The lang attribute of the content elements is used to specify the (human) language, allowing for correct representation of multi-lingual calibration certificates.

1.3. The DCC as a part of a digital laboratory

The strength of a digital calibration certificate lies in its ability to transfer calibration information in a machine-interpretable format. To take full advantage of this strength, the DCC should be embedded in the digital ecosystems of the issuing laboratory and the customer company.

National Metrology Institutes (NMIs), Designated Institutes (DIs), and other calibration laboratories issuing DCCs require middleware that supports the creation of DCCs and acts as an interface between analysis software, laboratory information management system (LIMS), and other electronic record keeping systems to collect all necessary information. With such systems in place, we foresee further automation of the creation of digital calibration certificates, reducing human error such as typos or copy-and-paste errors. PTB is currently developing such an interface (Oppermann et al., 2023), with a fully digital workflow already being piloted by a few select working groups within PTB (Keidel et al., 2025). NMIs and DIs will also benefit from automation of key comparisons and other inter-laboratory comparisons, where measurement results may be transmitted by the participants in DCC format for automated evaluation (see e.g. Rodiek et al. (2024)).

Customer laboratories should ideally process DCCs upon receipt, incorporating the transmitted information directly into their LIMS and other systems as appropriate. This processing should be automated to minimise human error. The flexible format allows for the inclusion and retrieval of meta data, e.g., information about measurement devices and standards used during calibration, in the DCC in a machine-interpretable format. Such information is useful in calculating correlations further along the calibration chain. The open-source electronic format also allows for secure long-term storage, including off-site backups, and future retrieval of digital calibration certificates, without the need to dedicate space and resources to long-term storage of paper calibration certificates. This is especially important in the field of radioactivity, where calibration standards may be in use for decades.

2. Towards a DCC for radioactivity

There are several steps to implementing DCCs for a new calibration community. First, one must collect examples for existing calibration certificates and applicable norms, and identify community-specific information that needs to be represented in a DCC. From this, one must then prepare two things: A list of refTypes (with definitions and usage notes) to encode the information in a machine interpretable way in the DCC, and a recommendation for how to structure the DCC (for example by means of a set of annotated good practice examples). Ideally, all this would be documented in a DKD expert report or similar publication.

The authors of this contribution are in the process of implementing DCCs for radioactivity, that is, the calibration of radionuclide standards with calibrated activity (or activity concentration), and the calibration of detectors that measure activities in Becquerel and related quantities. Other areas of ionising radiation metrology use much different quantities and measurement setups and will not be considered here. The work

³ www.ptb.de/dcc/dcc.xsd, accessed April 2, 2025.

⁴ digilab.ptb.de/dkd/refType/vocab/, last accessed: D:20250911

⁵ wiki.dcc.ptb.de/en/gp_home, accessed April 2, 2025.

⁶ www.ptb.de/cms/en/metrological-services/dkd.html, accessed April 2, 2025.

⁷ www.w3.org/TR/xmldsig-core/, accessed April 2, 2025.

⁸ digilab.ptb.de/dkd/refType/vocab/, accessed April 2, 2025.

⁹ www.ptb.de/dcc/v3.4.0-rc.1/, accessed April 2, 2025.

¹⁰ www.bipm.org/en/cipm-mra/cipm-mra-documents/service-categories, accessed April 2, 2025.

described here is very much in progress — input and feedback from the community are very welcome and necessary at this stage. The refTypes and examples shown here should not be considered exhaustive, but rather at this stage only cover the calibration of radionuclide standards in the form of point sources and liquid solutions.

Listing 1: Abridged example of a calibration item description. Some elements with mandatory children have been left empty here for the sake of readability.

```
<dcc:item id="P123-Am241" refType="Y_liquidSample">
  <dcc:name/>
  <dcc:description/>
  <dcc:manufacturer/>

  <dcc:identifications>
    <dcc:identification refType="basic_serialNumber">
      <dcc:issuer>manufacturer</dcc:issuer>
      <dcc:value>BD-6047</dcc:value>
      <dcc:name><dcc:content lang="en">Serial no.</
        dcc:content></dcc:name>
    </dcc:identification>
  </dcc:identifications>

  <dcc:itemQuantities>
    <dcc:itemQuantity refType="Y_massSolution">
      <si:real>
        <si:value>2.0184</si:value>
        <si:unit>\gram</si:unit>
        <si:expandedUnc>
          <si:uncertainty>0.0017</si:
            uncertainty>
          <si:coverageFactor>2</si:
            coverageFactor>
          <si:coverageProbability>0.95</si:
            coverageProbability>
        </si:expandedUnc>
      </si:real>
    </dcc:itemQuantity>
    <dcc:itemQuantity refType="
      Y_sampleCarrierSolution" refId="
      carrierSolution_Am241">
      <dcc:noQuantity/>
    </dcc:itemQuantity>
    <dcc:itemQuantity refType="Y_sampleNuclide" refId
      ="nuclide_Am241">
      <dcc:noQuantity/>
    </dcc:itemQuantity>
    <dcc:itemQuantity refType="Y_sampleContainer"
      refId="AM05">
      <dcc:noQuantity/>
    </dcc:itemQuantity>
  </dcc:itemQuantities>
</dcc:item>
```

2.1. RefTypes and refType prefix

Ionising radiation metrology, which includes the BIPM service categories of dosimetry, radioactivity, and neutron measurements, is part of DKD TC 2, “high frequency and optics”. The structure of refTypes and refType prefixes for the fields of metrology included in TC 2 are currently under discussion. In this text, we use the symbol “Y_” as a placeholder for the future refType prefix that will cover the new refTypes for radioactivity. All refTypes with the “Y_” prefix used in this document should be considered as preliminary, as the final list of refTypes is still being discussed within TC 2.

2.2. Calibration item

Calibration items in the field of radioactivity comprise devices (e.g. radon monitors, contamination monitors) and artefacts (radionuclide sources). Radionuclide sources are available in different shapes including point-like sources, area sources, liquid solutions, volume samples in complex matrices, and gaseous samples. In this contribution, we focus on the simplest case: A single-nuclide radioactivity standard provided in the form of a liquid solution, characterised in terms of its activity. Point sources and mixture sources easily generalise from the examples shown here. Area and volume sources are less common and their calibration certificates typically more complex as they must also be characterised in terms of their homogeneity. Gaseous sources are only relevant for a handful of nuclides. These cases, as well as DCCs for devices measuring radioactivity, will be considered in future works.

The definition of the calibration item(s) is part of the administrative data section of the DCC. In general, a DCC may have multiple calibration items (e.g., a set of weights). In the field of radioactivity, each calibration certificate commonly only covers a single calibration item. Still, we propose to assign an id attribute to the calibration item, to keep this structure as general as possible.

An example for a calibration item element for a liquid source can be seen in Listing 1. We propose at least the following information to be included:

- An id attribute to uniquely identify the calibration item within the DCC.
- A refType attribute specifying the type of sample.
- Child elements name, description, and manufacturer filled according to the DCC schema (not shown).
- The identifications element should contain at least one identification with refType basic_serialNumber, with a value corresponding to the serial number that the physical source is labelled with. Further identifiers may be listed.
- Physical properties of the calibration item of interest to the customer are listed under the itemQuantities element. These depend on the source type. For liquid sources (cf. Listing 1), we propose to list at least the mass and composition of the carrier solution, the radionuclides contained in the sample, and information about the sample container, each identified by a dedicated refType. Simple physical quantities, such as the mass of the sample solution, can be listed immediately, using the embedded D-SI schema. We propose to specify complex properties, such as the composition of the solution, within the statements element, and use the refId formalism to refer to them from the calibration item description.

Listing 2: Example of a radionuclide definition.

```
<dcc:list refType="Y_nuclideDefinitions">
  <dcc:list id="nuclide_Am241" refType="Y_nuclide">
    <dcc:name>
      <dcc:content lang="en">Americium-241 (Am-241)</
        dcc:content>
    </dcc:name>
    <dcc:quantity refType="basic_InChI">
      <dcc:noQuantity>
      <dcc:content>1S/Am/i1-2</dcc:content>
    </dcc:noQuantity>
    <dcc:quantity refType="Y_recommendedHalfLife">
      <si:real>
        <si:value>13651526177</si:value>
        <si:unit>\second</si:unit>
        <si:expandedUnc>
          <si:uncertainty>37868312</si:uncertainty>
          <si:coverageFactor>2</si:coverageFactor>
        </si:expandedUnc>
      </si:real>
    </dcc:quantity>
  </dcc:list>
</dcc:list>
```

```

    <si:coverageProbability>0.95</si:
      coverageProbability>
  </si:expandedUnc>
</si:real>
<dcc:measurementMetaData>
  <dcc:metaData>
    <dcc:reference>Evaluation: C.D. NESARAJA
      Publication cut-off: 30-Sep-2015 ENSDF
      insertion: 2015-12 Publication: Nuclear
      Data Sheets 130, 183 (2015)</dcc:
        reference>
    </dcc:metaData>
  </dcc:measurementMetaData>
</dcc:quantity>
<dcc:quantity refType="Y_isomericState">
  <dcc:charsXMLList>5/2 - 0</dcc:charsXMLList>
</dcc:quantity>
</dcc:list>
</dcc:list>

```

Listing 3: Example of a definition of a carrier solution

```

<dcc:list refType="Y_chemicalComponents">
  <dcc:quantity id="reagent_H2O" refType="
    Y_chemicalComponent">
    <dcc:name>
      <dcc:content lang="en">Water</dcc:content>
    </dcc:name>
    <dcc:noQuantity refType="basic_InChI">
      <dcc:content>1S/H2O/h1H2</dcc:content>
    </dcc:noQuantity>
  </dcc:quantity>
  <dcc:quantity id="reagent_HNO3" refType="
    Y_chemicalComponent">
    <dcc:name>
      <dcc:content lang="en">Nitric acid</dcc:content>
    </dcc:name>
    <dcc:noQuantity refType="basic_InChI">
      <dcc:content>1S/HNO3/c2-1(3)4/h(H,2,3,4)</dcc:
        content>
    </dcc:noQuantity>
  </dcc:quantity>
  <dcc:quantity id="carrier_LaNO33" refType="
    Y_chemicalComponent">
    <dcc:name>
      <dcc:content lang="en">Lanthanum(III) nitrate (La(
        NO3)3)</dcc:content>
    </dcc:name>
    <dcc:noQuantity refType="basic_InChI">
      <dcc:content>1S/La.3HNO3/c;3*2-1(3)4/h;3*(H
        ,2,3,4)/q+3;;</dcc:content>
    </dcc:noQuantity>
  </dcc:quantity>
</dcc:list>
<dcc:list id="carrierSolution_Am241" refType="Y_solution"
  >
  <dcc:list refType="Y_constituents">
    <dcc:quantity refType="Y_solvent" refId="reagent_H2O"
      >
      <dcc:noQuantity/>
    </dcc:quantity>
    <dcc:quantity refType="Y_molarConcentration" refId="
      reagent_HNO3">
      <si:real>
        <si:value>0.1</si:value>
        <si:unit>\mole\litre\tothe{-1}</si:unit>
      </si:real>
    </dcc:quantity>
    <dcc:quantity refType="Y_carrier Y_massConcentration"
      refId="carrier_LaNO33">
      <si:real>

```

```

    <si:value>35</si:value>
    <si:unit>\milli\gram\litre\tothe{-1}</si:unit>
    </si:real>
  </dcc:quantity>
</dcc:list>
</dcc:list>

```

Listing 4: Example of a container definition

```

<dcc:list id="AM05" refType="Y_container">
  <dcc:name>
    <dcc:content lang="en">Glass flask AM05</dcc:content>
  </dcc:name>
  <dcc:quantity>
    <dcc:noQuantity/>
    <dcc:measurementMetaData>
      <dcc:metaData>
        <dcc:reference>https://www.ptb.de/cms/ptb/
          fachabteilungen/abt6/fb-61/611-
          aktivitaetseinheit/abgabe-von-
          aktivitaetsnormalen/aktivitaetsnormale-in-
          form-von-loesungen.html</dcc:reference>
      </dcc:metaData>
    </dcc:measurementMetaData>
  </dcc:quantity>
</dcc:list>

```

2.3. Further definitions

The statements element within the administrative data section may contain general statements about the calibration laboratory or calibration certificate. This is also the element where further detailed information shall be placed that can then be referenced from elsewhere in the DCC. We propose to place definitions of complex properties of the calibration item here.

All radionuclides referenced in the results section should be listed, even if they were not detected in the sample. We propose to identify nuclides unambiguously by their InChI string (Heller et al., 2013) and isomeric state as given by the *J* (nuclear spin), *π* (parity), and *n* (energy level) quantum numbers. An example is given in Listing 2. Additional information of interest to the customer may be listed here, such as the half-life to be used when converting the activity to a different reference time. Another option currently under consideration is to reference an external database of radionuclides.

The proposed format for the description of the carrier solution for liquid samples is shown in Listing 3, where first the components/reagents are defined by their InChI strings. A similar format is foreseen to be used for other materials of interest, such as the foils or metal sheets that point sources and area sources adhere to.

We foresee some flexibility in the definition of the sample container. It will likely not be possible to define the shape, size, and material(s) in a fully machine-readable manner. However, this information may not be of interest to the customer anyway. In the example shown in Listing 4, we define the glass ampule by a reference to the calibration laboratory's web page, in accordance with the information given in analogue calibration certificates.

2.4. Measurement results

For a single-nuclide radioactivity standard, the quantity of interest is typically the activity at a given reference time. This quantity will be listed in the measurement results section in the form of an *si:real* quantity, whose child elements contain value, unit, and measurement uncertainty. It is important that the activity value be

Listing 5: Abridged example of a measurement result for single-nuclide sample characterised in terms of its activity. The uncertainty and list of equipment are not displayed here for conciseness.

```
<dcc:measurementResults>
  <dcc:measurementResult>
    <dcc:usedMethods>
      <dcc:usedMethod>
        <dcc:name>
          <dcc:content lang="en">HPGe gamma
            spectrometry</dcc:content>
        </dcc:name>
      <dcc:description/>
    <dcc:usedMethodQuantities>
      <dcc:usedMethodQuantity>
        <dcc:name/>
        <dcc:description/>
        <dcc:noQuantity refType="
          Y_measurementMethod">
          <dcc:content>UA-GR-GH
            -00-00-00</dcc:content>
        </dcc:noQuantity>
      </dcc:usedMethodQuantity>
    </dcc:usedMethodQuantities>
  </dcc:usedMethods>
</dcc:usedMethods>
<dcc:measuringEquipments/>
<dcc:name>
  <dcc:content lang="en">Activity measurement</
    dcc:content>
</dcc:name>
<dcc:results>
  <dcc:result refId="nuclide_Am241">
    <dcc:name>
      <dcc:content lang="en">activity of Am
        -241</dcc:content>
    </dcc:name>
    <dcc:data>
      <dcc:quantity refType="
        basic_measuredValue Y_activity"
        refId="nuclide_Am241">
        <si:real>
          <si:value>2.52</si:value>
          <si:unit>\mega\becquerel</si:
            unit>
          <si:dateTime>2022-08-01T13
            :00:00+01:00</si:dateTime
              >
          <si:expandedUnc>...</si:
            expandedUnc>
        </si:real>
      </dcc:quantity>
    </dcc:data>
  </dcc:result>
</dcc:results>
</dcc:measurementResult>
</dcc:measurementResults>
```

unambiguously associated to the reference time for which it is valid. (The exception would be long-lived radionuclides with half-lives in excess of the expected duration of human civilisation, as their activities can be considered constant in the context of calibration sources). We propose to list the reference time in the `si:dateTime` child element of `si:real`. The element `si:dateTime` contains a time stamp in ISO 8061 format [ISO Central Secretary \(2019\)](#), with the timezone defined by its offset to UTC.

An example for a single-nuclide measurement result is shown in Listing 5. The refTypes `basic_measuredValue` and `Y_activity`,

in combination with the reference back to the previously defined `nuclide_Am241`, identify this result as the measured activity of Am-241, with value 2.52 MBq at 1 PM Central European Time (UTC+1) on August 1, 2022. For mixture samples comprising multiple radionuclides, we propose to add one `result` element per nuclide within the same `measurementResult` element.

For independent measurements (e.g., an area source that is characterised both in terms of its activity and its homogeneity), we propose a second `measurementResult` element be added.

The DCC schema allows for the inclusion of meta data at many levels of the measurement results section. Listing 5 shows, for example, where information about the measurement method(s) and equipment may be included. These are not filled in the example. The DCC schema specifies a general structure for each of the metadata elements. However, the details for how to implement information about e.g. the methods in a manner that would be machine readable in the field of radionuclide metrology is still being worked out. We suggest to include the acronyms defined by BIPM CCRI(II)¹¹ in some way.

We propose to define refTypes for all measurement quantities needed for radionuclide metrology calibration certificates, such as activity, activity per mass, activity per volume, and activity ratios. A full list of proposed refTypes is in preparation. The refType definition and usage notes should specify how the reference time and radionuclide should be included.

We also propose to define refTypes for the decision threshold and limit of detection, to quantify the absence of impurities. These would be used in conjunction with a refType defining a quantity, such as `Y_activity`.

3. Extensions and future work

3.1. Extended measurement data

There are situations in which one might want to provide additional information to the customer, or one might want to provide calibration information and additional measurements outside of a traditional calibration certificate. This could include situations such as inter-laboratory comparisons, where multiple laboratories characterise the same radiation source, or scientific data exchange. The DCC schema already includes the option to specify a variety of metadata. In addition to information on the measurement methods and equipment, one may report software used in the measurement or data evaluation, ambient conditions such as the laboratory temperature or other influence conditions, or general metadata.

We are planning to provide examples on how the results typically provided by participants in inter-laboratory comparison reports may be included in a DCC. We have also developed an example for the inclusion of gamma-ray spectrometry measurements, which may be generalised to any kind of measurement data.

3.2. Towards a fully digital key comparison

A Digital Metrological Expert (DME) is a tool able to perform analyses of and report on inter-laboratory comparisons with minimal intervention from human subject matter experts, as long as each lab-

¹¹ www.bipm.org/documents/20126/48200451/working-document-ID-1531/63c860ea-9b47-41fd-f9c2-50bda1fdbb44, accessed April 4, 2025.

oratory provides results in a suitable digital format. At least one such DME is currently being developed (Rodiek et al., 2024). In the future, we plan to show a proof-of-principle of how such a DME might interface with an inter-laboratory comparison in the field of radionuclide metrology.

3.3. Time-dependent activity and uncertainty

The activity of radioactive sources changes over time due to the nature of radioactive decay. Current best practice is to provide the activity (including uncertainty) at a reference time (usually specified by the customer), and leave it to the customer to perform any desired conversions to other times and dates as needed. This may be combined with statements about the value of the half-life of the nuclide and/or the state of equilibrium if multiple radionuclides in the same decay chain have to be considered. This approach requires some knowledge by the customer about the physics of nuclear decays, namely knowledge of decay laws and when a “simple” exponential decay may be assumed.

At this time, the DCC already supports the inclusion of equations/formulas, both in human-readable and machine-readable formats, for example using the mathML standard (ISO Central Secretary, 2016). There is still ongoing work in the community on how the formula constituents can best be associated to other elements of the DCC, so that their meaning and interpretation is stored in a machine-readable fashion. Once this has been settled, we propose to include the equations needed to convert measured activities to a different reference time in the DCC, to allow software to perform this calculation independently of human subject matter experts.

4. Summary and conclusions

Digital calibration certificates will provide a more streamlined, less error prone way of disseminating calibration results to calibration laboratories and end users. We have shown here how different aspects of radionuclide metrology (description of the calibration item, measurement results) may be implemented as a DCC. This is a work in progress — we welcome feedback and participation from the community. A full DKD expert report, which will contain a list of refTypes, is in preparation.

CRedit authorship contribution statement

Henrike Fleischhack: Conceptualization, Methodology, Investigation, Writing – original draft. **Matthias O. Stein:** Conceptualization, Methodology, Writing – review & editing. **Stefan Röttger:** Supervision, Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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